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Industrial Internship Report on

” Banking Information System”

Prepared by

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Executive Summary
My project was Banking Information System. It is a Java-based application that allows users to perform banking operations such as deposit, withdrawal, and balance enquiry. **3. Problem Statement** **The problem is to design a simple banking system that allows users to manage their bank accounts digitally and perform transactions efficiently.** **4. Existing vs Proposed Solution** **Existing:** • **Manual banking system** • **Time consuming** **Proposed:** • **Automated Java system** • **Fast and accurate**

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1 Preface

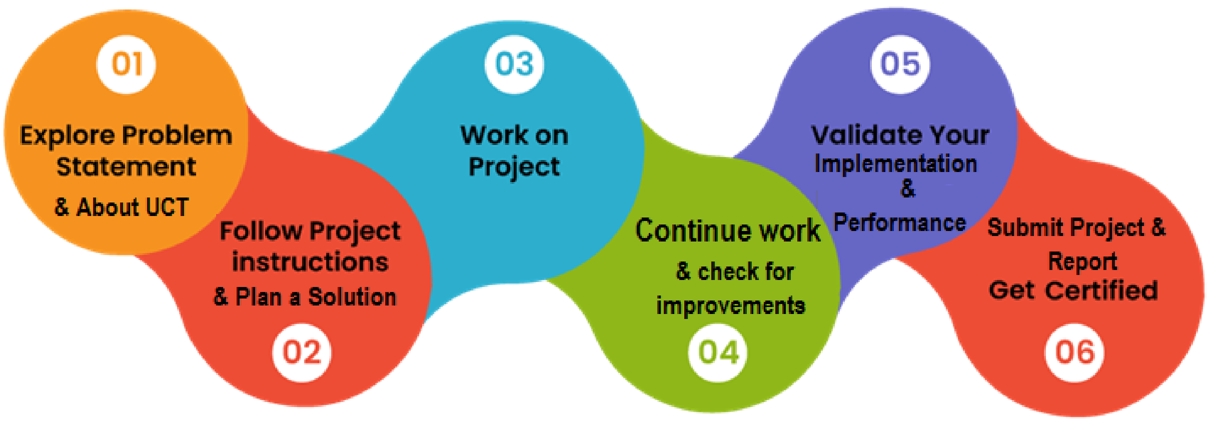
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

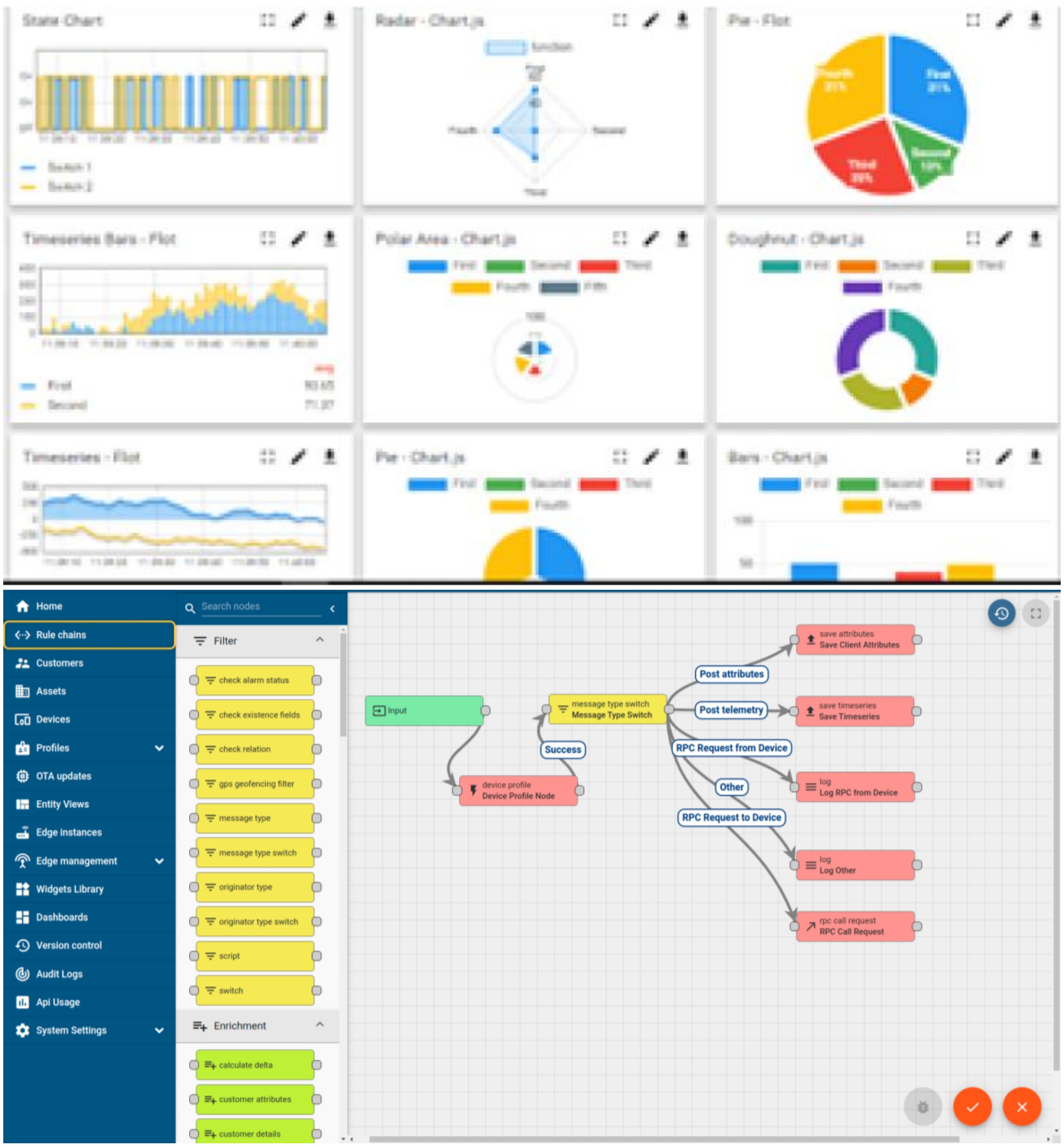
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

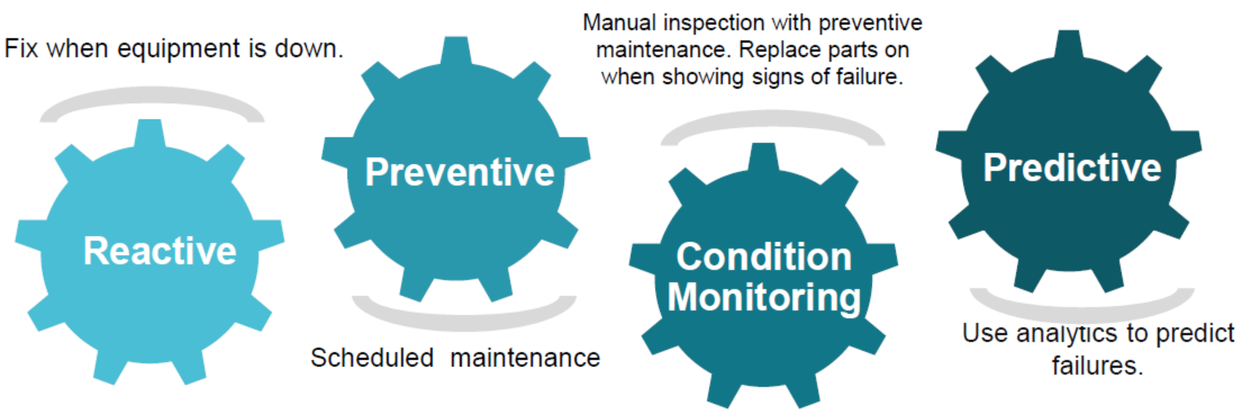
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

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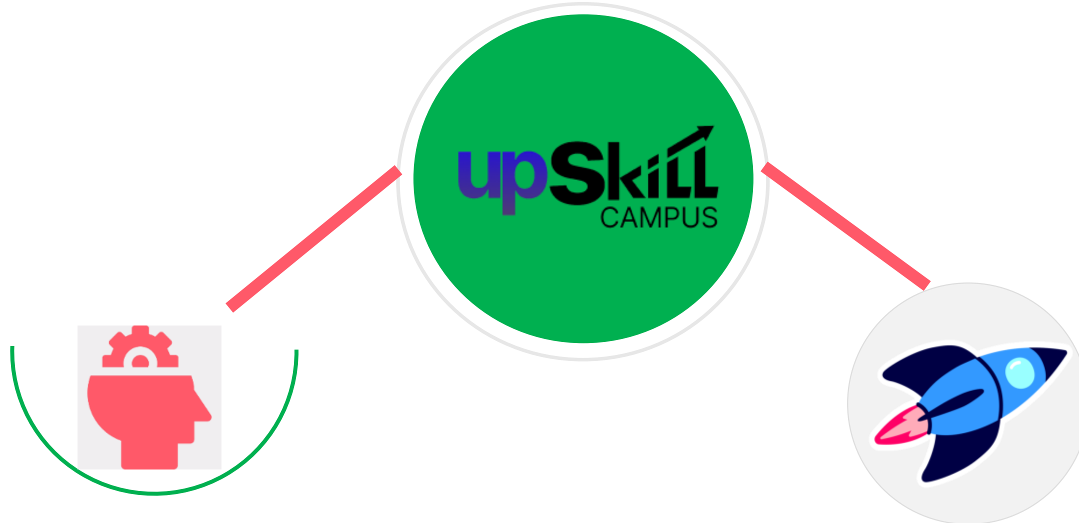
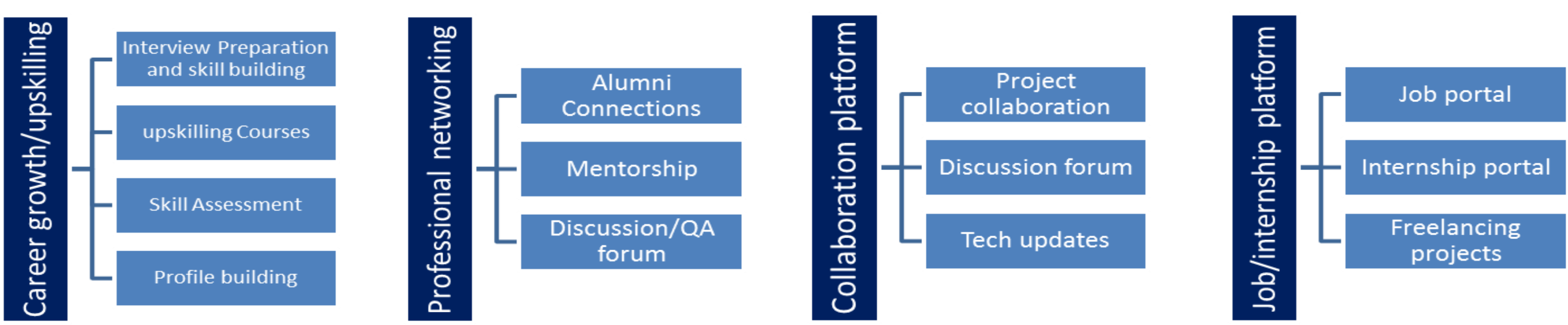


Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

3 Problem Statement

In the assigned problem statement

The assigned problem is to design and develop a **Banking Information System** that can perform basic banking operations in an efficient and user-friendly manner. Traditional manual banking processes are time-consuming, prone to errors, and lack quick accessibility for users.

The objective of this project is to create a Java-based application that allows users to:

- Create and manage bank accounts
- Deposit money
- Withdraw money
- Check account balance

The system should ensure accuracy in transactions, provide quick responses, and reduce human errors. It should also be simple to use and demonstrate the basic functionality of a real-world banking system.

4 Existing and Proposed solution

- **Existing Solutions**

In traditional banking systems, many operations are handled manually or through basic software systems. These systems often require physical presence at the bank and involve paperwork for transactions like deposits and withdrawals.

Some limitations of existing solutions include:

- Time-consuming processes
- Higher chances of human error
- Limited accessibility for users
- Lack of automation in small-scale systems
- Dependency on manual record maintenance

- **Proposed Solution**

The proposed system is a Banking Information System developed using Java. It is a simple console-based application that automates basic banking operations.

The system allows users to:

- Deposit money
- Withdraw money
- Check account balance

All operations are performed digitally within the program, ensuring faster execution and improved accuracy.

4 Value Addition

The proposed system adds value in the following ways:

- Reduces manual effort and errors
- Provides quick and accurate transaction results
- Easy to use and understand

- Demonstrates real-world banking functionality
- Can be further extended with database and GUI features

4.1 Code submission (Github link)

https://github.com/SanchanaaSrinath/upskillcampus/blob/main/BankingInformationSystem.java

4.1 Report submission (Github link) :

https://github.com/SanchanaaSrinath/upskillcampus

Proposed Design/ Model

The proposed Banking Information System is designed as a simple console-based application using Java. The system follows a structured flow consisting of input, processing, and output stages.

At the start, the program initializes the account balance and displays a menu to the user. The user selects an operation such as deposit, withdrawal, or balance enquiry. Based on the selected option, the system processes the request using conditional statements and updates the account balance accordingly.

The system ensures that withdrawal operations are performed only when sufficient balance is available. After each operation, the updated balance is displayed to the user. This process continues until the user chooses to exit the program.

The design is simple, efficient, and easy to understand, making it suitable for demonstrating real-world banking operations in a basic form.

4.2 High Level Diagram

User → Input → Java Program → Processing → Output

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

4.3 Low Level Diagram

Start

↓

Display Menu

↓

User Choice

↓

Processing (Deposit / Withdraw / Check Balance)

↓

Update Balance

↓

Display Result

↓

Repeat or Exit

4.4 Interfaces

The Banking Information System uses a simple console-based interface to interact with the user. The system is designed in a structured manner to handle input, processing, and output efficiently.

Block Diagram

User → Input Interface → Processing Unit → Output Interface

Data Flow

- User enters data
- System processes request
- Balance is updated
- Result is displayed
- **Protocols**
- **No external communication protocols are used**
- **The system runs locally using Java**
- **Input/output handled via standard console**

- **State Management**

The system works based on simple states:

- **Idle State → Waiting for user input**
- **Processing State → Performing operation**
- **Output State → Displaying result**

- **Memory Management**
- **Balance stored in a variable (double balance)**
- **Temporary variables used for transactions**
- **No external database used**
- **Efficient memory usage as program is lightweight**

5 Performance Test

The performance test evaluates how effectively the Banking Information System works under different conditions and constraints. Even though the system is a simple console-based application, it is designed to simulate real-world banking operations efficiently.

- **Constraints Identified**

The following constraints were considered during the development of the system:

- **Speed (Execution Time):** The system should respond quickly to user inputs
- **Accuracy:** Transactions like deposit and withdrawal must be correct
- **Memory Usage:** The program should use minimal memory
- **Usability:** The system should be simple and easy to use

- **Handling of Constraints**
- The program uses simple and efficient Java logic (switch-case, variables), ensuring fast execution

- Accuracy is maintained by validating withdrawal conditions (checking sufficient balance)
- Memory usage is minimal since only basic variables are used (no database or large storage)
- A menu-driven interface improves usability and reduces user errors

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- **Test Plan / Test Cases**

Some test cases used to evaluate the system:

- Deposit valid amount → balance increases correctly
- Withdraw valid amount → balance decreases correctly
- Withdraw more than balance → error message displayed
- Check balance → correct value displayed
- Invalid choice → handled properly

- **Test Procedure**
- The program was executed multiple times with different inputs
- Various transaction scenarios were tested
- Edge cases like insufficient balance were verified

- **Performance Outcome**
- The system executed all operations quickly without delay
- All transactions were accurate
- No memory issues were observed
- The system handled invalid inputs effectively

- **Conclusion**

The Banking Information System performs efficiently under the defined constraints. Although it is a basic implementation, it demonstrates the core principles of performance, accuracy, and usability required in real-world applications. With further enhancements like database integration and GUI, the system can be scaled for industrial use.

4.5 Test Plan/ Test Cases

The test plan is designed to verify the correctness, reliability, and functionality of the Banking Information System. Different test cases were executed to ensure that all operations work as expected.

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5 Test Cases

Test Case ID	Description	Input	Expected Output	Result
TC01	Deposit valid amount	1000	Balance updated to 1000	Pass
TC02	Withdraw valid amount	500	Balance reduced correctly	Pass
TC03	Withdraw more than balance	2000	“Insufficient Balance” message	Pass
TC04	Check balance	—	Displays current balance	Pass
TC05	Invalid menu choice	9	“Invalid Choice” message	Pass
TC06	Multiple transactions	Deposit		

5.1 Test Procedure

The testing of the Banking Information System was carried out by executing the program multiple times with different inputs and scenarios.

- The application was run in a Java environment
- Each menu option was selected and tested individually
- Valid and invalid inputs were entered to check system response
- Boundary conditions such as insufficient balance were tested
- The output was observed and compared with expected results

The system was tested repeatedly to ensure consistent performance and correct functionality of all operations.

5.2 Performance Outcome

The Banking Information System demonstrated efficient performance during testing. All operations such as deposit, withdrawal, and balance enquiry were executed quickly and accurately.

The system responded instantly to user inputs, indicating good execution speed. The accuracy of transactions was maintained throughout, as the balance was correctly updated after each operation. Error handling was effective, especially in cases like insufficient balance and invalid menu choices.

The application consumed very minimal memory since it uses simple variables and does not involve complex data structures or databases. The system remained stable during multiple test runs and did not crash or produce unexpected results.

Overall, the performance of the system meets the basic requirements of a real-world application in terms of speed, accuracy, and reliability. It provides a strong foundation that can be further enhanced for large-scale or industrial use.

6 My learnings

During this internship, I gained valuable knowledge and practical experience in developing a real-world application. I learned how to design and implement a **Banking Information System using Java**, which helped me strengthen my programming skills.

I improved my understanding of core Java concepts such as variables, control structures (if-else, switch-case), loops, and user input handling using the Scanner class. I also learned how to structure a program logically and handle different scenarios like valid and invalid inputs.

This project enhanced my problem-solving abilities and helped me understand how real-world systems are designed and developed. I also gained experience in using GitHub for code submission and version control, which is an important skill in the software industry.

Overall, this internship helped me build confidence in coding, improved my technical skills, and gave me a clear understanding of how software development works in an industry environment. These learnings will be very useful for my future career growth in the field of software development

7 Future work scope

The current Banking Information System is a basic console-based application. There are several improvements that can be made in the future to enhance its functionality and make it more suitable for real-world applications.

Some possible future enhancements include:

- Developing a **Graphical User Interface (GUI)** to make the system more user-friendly
- Integrating a **database (MySQL)** to store user account details permanently
- Adding **user authentication (login system)** for security
- Implementing additional features such as fund transfer, transaction history, and account management
- Enhancing security by adding encryption and validation mechanisms
- Converting the system into a **web-based application** using full stack technologies

These improvements will make the system more scalable, secure, and efficient, bringing it closer to real-world banking applications.